

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | **Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU is exclusively English-language, text-only, multiple-choice [Q8, Q21, Q45, Q46]. The HuggingFace metadata confirms monolingual English; every one of the 175+ examples sampled is English-only [DATASET-D2, D29, etc.]. The deployment elicitation explicitly requires Telugu, Bangladeshi Bengali (Mymensingh dialect), and Indian Bengali as query languages [elicitation A4, Q4]. Bengali script (বাংলা) and Telugu script (తెలుగు) are entirely unsupported. The paper's own authors acknowledge the limitation that 'many important concepts are conveyed mainly through other modalities' [Q64]. No transliteration, code-switching, or multilingual variant exists. Web research confirms no Bengali or Telugu agricultural-domain LLM evaluation infrastructure exists [WEB-5, WEB-17, WEB-18, WEB-25] and that LLMs perform meaningfully worse in Bengali than English [WEB-5]. This is an absolute, format-level mismatch that no in-format remediation can address. IF is a HIGH-priority dimension for this deployment, making this a fundamental validity violation.

ID	Evidence
Q4	"However, on every one of the 57 tasks, the best models still need substantial improvements before they can reach expert-level accuracy." (p.1)
Q8	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q21	"We collected 15908 questions in total, which we split into a few-shot development set, a validation set, and a test set. The few-shot development set has 5 questions per subject, the validation set may be used for selecting hyperparameters and is made of 1540 questions, and the test set has 14079 questions. Each subject contains 100 test examples at the minimum, which is longer than most exams designed to assess people." (p.3)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with 'The following are multiple choice questions (with answers) about [subject].' For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with 'Answer: '. The model then produces probabilities for the tokens 'A,' 'B,' 'C,' and 'D,' and we treat the highest probability option as the prediction." (p.6)
Q46	"For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q64	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
D2	"The song, and the movement that it was connected to, highlight which of the following developments in the broader society in the late 1800s?" (re: "The Hayseed", US Populist movement) — US-specific historical content with no South Asian relevance
D29	If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel? — US units (miles per hour) — not metric; minor cultural assumption
WEB-5	arXiv 2507.23248 — LLMs perform worse in Bengali than English; smaller models especially degraded (https://arxiv.org/pdf/2507.23248)
WEB-17	AI4Bharat IndicBERT covers Bengali and Telugu but not agricultural domain (https://github.com/AI4Bharat/IndicBERT)
WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
WEB-25	IndicXTREME (ACL 2023) covers Bengali/Telugu for general NLU but no agricultural evaluation (https://aclanthology.org/2023.acl-long.693)

Strengths

- Standardized text-MCQ format is portable and computationally tractable for cross-model comparison [Q23, Q35]
- Format insensitivity findings for GPT-3 [Q100] mean that translated/adapted versions could in principle preserve relative model rankings if translation pipeline were added

ID	Evidence
Q23	"we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions." (p.3)
Q35	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q100	"While different question formatting choices often lead to similar GPT-3 accuracies, we find that UnifiedQA is more sensitive." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution mismatch is total: source benchmark is English ASCII text [Q8, Q45]; target deployment requires Bengali script (Bangladeshi and Indian variants), Telugu script, and English. No regional language input is supported.

IF-2: Regional infrastructure does support text input — Bangladesh has 44.5–80% internet penetration [WEB-13, WEB-14] and mobile-first usage [WEB-15] — but this is irrelevant since MMLU's English-only constraint is the binding one, not regional capture capacity.

IF-3: Domain-specific form differences include: (a) Bengali/Telugu script Unicode handling (no uppercase/lowercase distinction), (b) absence of multimodal capacity (acknowledged limitation [Q64]) which excludes diagrams/maps relevant to delta hydrology and AEZ classification, (c) no code-switching support relevant to Bengali-English bilingual scientific register.

IF-4: Form mismatch is absolute and constitutes a complete external validity violation for the multilingual deployment requirement; no within-benchmark workaround exists.

ID	Evidence
Q8	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q45	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction." (p.6)
Q64	"While text is capable of conveying an enormous number of concepts about the world, many important concepts are conveyed mainly through other modalities, such as images, audio, and physical interaction (Bisk et al., 2020). Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
WEB-5	arXiv 2507.23248 — LLMs perform worse in Bengali than English; smaller models especially degraded (https://arxiv.org/pdf/2507.23248)
WEB-13	https://datareportal.com/reports/digital-2024-bangladesh
WEB-14	https://www.newagebd.net/post/telecom/252740/internet-penetration-rate-declines
WEB-15	https://en.wikipedia.org/wiki/Internet_in_Bangladesh
WEB-17	AI4Bharat IndicBERT covers Bengali and Telugu but not agricultural domain (https://github.com/AI4Bharat/IndicBERT)

Information Gaps

- Mymensingh-dialect-specific phonological/lexical variants in agricultural register are not documented in any retrievable NLP corpus and would require in-region linguist elicitation

Remediation

Gap 1: English-only text input precludes Bengali (both variants), Telugu, and any multilingual evaluation explicitly required by the deployment.

Recommendation: Do not use MMLU for any multilingual capability assessment. Build a multilingual evaluation pipeline using IndicXTREME [WEB-25] as the structural template and IndicTrans2 [WEB-18] for translation infrastructure, with native-script Bengali and Telugu items. Plan evaluation to detect language-mismatch failures (e.g., model responding in English to Bengali prompts).

ID	Evidence
WEB-18	IndicTrans2 covers 22 Indian languages including Telugu but is a translation system, not knowledge benchmark (https://openreview.net/forum?id=vfT4YuzAYA)
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